

■ AgriTrust Community Credit Network

Complete Technical Platform Documentation & Source Code Guide

Architecture: React 18 + Vite Express.js PostgreSQL	Security: Unique Farmer Isolation & ZK Consent Passport
ML Modules: Credit Engine, AI Repayment Risk, JLG Credit	Version: 1.0.0 Production Enterprise Release

1. Executive Summary & Core Innovations

AgriTrust Community Credit Network is an alternative credit-scoring, dynamic risk evaluation, and community lending platform built for agricultural micro-finance and rural banking.

Key Innovations:

- **Alternative Data Credit Scoring:** Replaces paper collateral with multi-source verified data (PM-KISAN DBT, ICAR satellite canopy health, NHB yield data, AGMARKNET mandi prices).
- **AI Continuous Repayment Risk Prediction:** Dynamically evaluates farmer risk on a 0–100 scale (LOW/MEDIUM/HIGH) based on Debt Service Coverage Ratios (DSCR), weather, and mandi volatility.
- **Joint Liability Group (JLG) Credit:** Enables 3 to 10 farmers to pool credit limits with a joint guarantee drawdown mechanism.
- **Strict User Data Isolation:** Newly registered farmers receive a unique ID (FARM-XX-XXXX) and start 100% clean without data bleed from existing users.

2. Technical Stack & Directory Structure

Layer	Technology	Description
Frontend	React 18 + Vite 6	Single-Page App with Tailwind CSS & Lucide icons
Charts	Recharts	Dynamic risk breakdown & credit trend charts
Backend	Node.js + Express 4	REST APIs, auth, credit engines & audit logging
Database	PostgreSQL 16 + JSON Persistence	Relational DB with JSON persistence mode

```
c:\Users\ktanp\OneDrive\Desktop\IBM
■■■ package.json           # Dependencies & scripts
■■■ vite.config.js         # Vite configuration
■■■ tailwind.config.js     # Styling config
■■■ src/
■   ■■■ App.jsx            # Main Layout & Tab Router
■   ■■■ components/
■       ■■■ LoginPage.jsx  # Authentication & Registration
■       ■■■ FarmerVault.jsx # Farmer Dashboard & Edit Form
■       ■■■ FPOCommunityHub.jsx # FPO & JLG Group Hub
■       ■■■ BankCreditPortal.jsx # Bank Underwriting Portal
■       ■■■ AdminUsersAudit.jsx # Platform Admin & Audit Table
■       ■■■ FarmerHistoryModal.jsx # Farmer Profile & Audit Trail Modal
■       ■■■ GroupHistoryModal.jsx # Group History & Audit Trail Modal
■   ■■■ utils/
■       ■■■ reportGenerator.js # Client PDF Report Generator
■■■ server/
■   ■■■ index.js           # Express Backend API Engine
■   ■■■ db.js              # PostgreSQL & Persistence Layer
■   ■■■ services/
■       ■■■ creditScoringEngine.js # Credit Scoring Engine
■       ■■■ repaymentRiskAI.js    # AI Risk Prediction Engine
■       ■■■ groupCreditEngine.js  # JLG Group Credit Engine
■       ■■■ farmerHistoryService.js # Farmer History Aggregator
■       ■■■ groupHistoryService.js # Group History Aggregator
■       ■■■ auditLogger.js        # Audit Logger Service
```

3. Package Dependencies (`package.json`)

```
{
  "name": "agritrust-community-credit-network",
  "private": true,
  "version": "1.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview",
    "server": "node server/index.js",
    "start": "concurrently \"npm run server\" \"npm run dev\""
  },
  "dependencies": {
    "@radix-ui/react-accordion": "^1.2.20",
    "@radix-ui/react-dialog": "^1.1.23",
    "cors": "^2.8.5",
    "express": "^4.21.2",
    "lucide-react": "^0.469.0",
    "pg": "^8.13.1",
    "react": "^18.3.1",
    "react-dom": "^18.3.1",
    "recharts": "^2.15.0",
    "tailwind-merge": "^3.6.0"
  },
  "devDependencies": {
    "@vitejs/plugin-react": "^4.3.4",
    "autoprefixer": "^10.4.20",
    "concurrently": "^9.1.2",
    "tailwindcss": "^3.4.17",
    "vite": "^6.0.7"
  }
}
```

4. Backend API Routes & Authentication (`server/index.js`)

- **POST /api/auth/login:** Authenticates farmer accounts by Mobile No or Farmer ID. Scopes returned payload strictly to the matched farmer ID.
- **POST /api/auth/register-farmer:** Assigns a unique FARM-XX-XXXX ID, initializes clean empty peer and acreage records, and logs a FARMER_REGISTERED audit event.
- **PUT /api/farmer/update-profile:** Updates farmer details safely without default landholding overrides.
- **GET /api/history/farmer/:farmerId:** Aggregates personal info, credit scores, loan history, repayment defaults, AI risk predictions, and complete audit logs.
- **GET /api/history/group/:groupId:** Aggregates JLG group trust score history, member sub-limits, and drawdown logs.

5. Quality Assurance & 10-Step Automated Verification

10-Step Automated Verification Test Result:

- ✓ Step 1: Registered Farmer A (FARM-MA-4961).
- ✓ Step 2: Verified Farmer A initial Edit Details form is 100% empty/zero (land: 0, crop: Not Specified, score: null).
- ✓ Step 3: Saved Farmer A details (Land: 4.5, Crop: Pomegranate, Peers: Alice & Bob).
- ✓ Step 4: Logged out Farmer A.
- ✓ Step 5: Registered Farmer B (FARM-MA-7316).
- ✓ Step 6: Verified Farmer B Edit Details form has **ZERO data bleed from Farmer A.**
- ✓ Step 7: Saved Farmer B details (Land: 2.0, Crop: Onion, Peer: Charlie).
- ✓ Step 8: Switched back to Farmer A and verified Farmer A retained **ONLY** Farmer A's saved details.
- ✓ Step 9: Verified existing legitimate demo farmers (FARM-MH-9103) retain historical data intact.
- ✓ Step 10: Production build test completed cleanly with **2206 modules transformed and 0 errors.**